

Yiran Pang

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Education

Florida Atlantic University, *Ph.D. Candidate in Computer Science*, GPA: 4.0 / 4.0

Sept 2022 – Present

- **Advisor:** Prof. Zhen Ni and Prof. Xiangnan Zhong
- **Research Area:** Federated Learning, Reinforcement Learning, Large Language Models, and Computer Vision
- **Teaching Assistant:** CAP6629: Reinforcement Learning (Spring/Fall 2025); COT6405: Analysis of Algorithms (Spring/Fall 2024); CAP5625: Computational Foundations AI (Summer 2025)

Publications

Y. Pang, Z. Ni, and X. Zhong, Decoupling Shared and Personalized Knowledge: A Dual-Branch Federated Learning Framework for Multi-Domain with Non-IID Data, *AAAI Conference on Artificial Intelligence (AAAI-26)*.

Y. Pang, Y. Zhao, Z. Zhou, T. Hu, and R. Hou, “Is OpenVLA Truly Robust? A Systematic Evaluation of Positional Robustness,” in *Proc. International Joint Conference on Natural Language Processing and Asia-Pacific Chapter of the Association for Computational Linguistics (IJCNLP–ACL)*, 2025.

Y. Pang, Z. Ni, and X. Zhong, “A fast federated reinforcement learning approach with phased weight-adjustment technique,” *Neurocomputing*, vol. 626, p. 129550, 2025.

Y. Pang, Z. Ni, and X. Zhong, “Personalized Observation Normalization for Federated Reinforcement Learning in Simulation Environments with Heterogeneity,” in *Proc. IEEE International Joint Conference on Neural Networks (IJCNN)*, 2025.

Y. Pang, Z. Ni, and X. Zhong, “Integration of a new layer normalization process into federated reinforcement learning for environments with heterogeneous attribute spaces,” in *Proc. SPIE Artificial Intelligence and Machine Learning for Multi-Domain Operations Applications VII*, vol. 13473, pp. 246–253, 2025.

X. Min, W. Zhou, R. Hu, Y. Wu, **Y. Pang**, and J. Yi, “Lwuavdet: A lightweight UAV object detection network on edge devices,” *IEEE Internet of Things Journal*, vol. 11, no. 13, pp. 24013–24023, 2024.

W. Zhou, F. Zheng, Y. Zhao, **Y. Pang**, and J. Yi, “MSDCNN: A multiscale dilated convolution neural network for fine-grained 3D shape classification,” *Neural Networks*, vol. 172, p. 106141, 2024.

Z. Wang, **Y. Pang**, Y. Lin, and X. Zhu, “Adaptable and reliable text classification using large language models,” in *Proc. IEEE Int. Conf. on Data Mining Workshops (ICDMW)*, pp. 67–74, 2024.

Z. Wang, **Y. Pang**, and Y. Lin, “Large language models are zero-shot text classifiers,” *arXiv preprint arXiv:2312.01044*, 2023.

Y. Pang, Z. Ni, and X. Zhong, “Federated learning for crowd counting in smart surveillance systems,” *IEEE Internet of Things Journal*, vol. 11, no. 3, pp. 5200–5209, 2023.

Z. Wang, **Y. Pang**, C. Ulus, and X. Zhu, “Counting manatee aggregations using deep neural networks and Anisotropic Gaussian Kernel,” *Scientific Reports*, vol. 13, no. 1, p. 19793, 2023.

J. Yi, **Y. Pang**, W. Zhou, M. Zhao, and F. Zheng, “A perspective-embedded scale-selection network for crowd counting in public transportation,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 25, no. 5, pp. 3420–3432, 2023.

W. Zhou, X. Min, Y. Zhao, **Y. Pang**, and J. Yi, “A multi-scale spatio-temporal network for violence behavior detection,” *IEEE Transactions on Biometrics, Behavior, and Identity Science*, vol. 5, no. 2, pp. 266–276, 2023.

W. Zhou, F. Zheng, G. Yin, **Y. Pang**, and J. Yi, “Yolotrashcan: A deep learning marine debris detection network,” *IEEE Transactions on Instrumentation and Measurement*, vol. 72, pp. 1–12, 2022.

Y. Pang, Crowd Density Analysis Method in Scenic Spot Based on Multi-Source Feature Fusion, *Master Thesis, Chongqing University of Science and Technology*, 2022.

Y. Wang, W. Zhong, H. Su, F. Zheng, **Y. Pang**, H. Wen, and K. Cai, “An improved MVCNN for 3D shape recognition,” in *Proc. IEEE Int. Conf. on Emergency Science and Information Technology (ICESIT)*, pp. 469–472, 2021.

► Submitted / Under Preparation:

Y. Pang, W. Cheng, Z. Ni, and X. Zhong, “S2RMaze: A Sim-to-Real Maze Navigation Dataset for Learning-Driven Methods via Mobile Robots,” *Under Preparation*, 2025.

Projects

Wearable Soft Magnetic Sensor Arrays for Robust Force Myogram Recognition in Prosthetic Hand Control 2025-Present

- Building upon a soft magnetic FMG prosthetic sensing system for real-time muscle pattern recognition
- Developing algorithms to address sensor displacement and subject variability caused by different wearing positions
- Exploring cross domain adaptation methods to improve robustness and classification accuracy

Intelligent Data Analysis and Decision System for Construction Sites 2022

- Designed real-time safety compliance inspection functions using YOLO for detecting unsafe behaviors and warning signs
- Developed server-side video scheduling and parallel detection interfaces for multi-stream processing and early warning storage

Anti-Forgery Code Identification System for Cigarette Packaging 2021

- Developed a 32-bit CRNN-based recognition network for cigarette packaging anti-counterfeiting code (laser codes on transparent packaging) up to commercial standards
- Migrated PyTorch models (CRNN, CTPN, YOLO, DBNet) to NCNN via ONNX; optimized for C++ execution and cross-platform deployment
- Integrated C++ and Java through JNI for Android app and WeChat mini program deployment

Recognition System for Traditional Chinese Medicine Pieces 2021

- Built a ResNet-based classification and quality grading model for traditional Chinese medicine pieces
- Collected and cleaned dataset; improved model accuracy from 79% to 96.9% via augmentation and refinement
- Deployed network service for efficient mobile access

Big Data Safety Monitoring and Management System for Tourism 2019

- Enhanced SlowFast network for real-time violence recognition, improving accuracy by 5%
- Improved crowd counting accuracy by 30% through algorithm optimization for surveillance data
- Developed abnormal crowd behavior detection and real-time alert functions for high-risk scenes

Awards and honors

2024	Best Journal Publication from FAU College of Engineering & Computer Science
2023-2025	Academic Achievement from FAU College of Engineering & Computer Science
2021	2 nd Prize of the National Mathematical Contest in Modeling
2021	3 rd Prize of the Graduate Electronic Design Contest
2020	3 rd Prize of the National Mathematical Contest in Modeling

Talks

- 2025 Yiran Pang, “A fast federated reinforcement learning approach with phased weight-adjustment technique,” Special Topics in Reinforcement Learning, CAP6629, FAU.
- 2025 Yiran Pang, “Introduction to Reinforcement Learning in Control Tasks,” Engineer the Future, FAU.
- 2025 Yiran Pang, “Personalized Observation Normalization for Federated Reinforcement Learning in Simulation Environments with Heterogeneity,” IJCNN (Oral).
- 2025 Yiran Pang, “Integration of a new layer normalization process into federated reinforcement learning for environments with heterogeneous attribute spaces,” Special Topics in Reinforcement Learning, CAP6629, FAU and SPIE (Oral).
- 2024 Yiran Pang, “Federated Learning for Crowd Counting in smart surveillance systems,” Academic Excellence Event, FAU.